LAB 5 – STATIC CODE ANALYSIS

NAME : L A GOKUL	SRN : PES2UG24CS246
SECTION : D	DATE : 14/10/2026
PROGRAM : WAREHOUSE STOCK MANAGER	SUBJECT : SOFTWARE ENGINEERING

1. Aim :

To check the warehouse stock manager program using Pylint, Bandit and Flake8, fix the issues reported by the tools, and check the program again.

2. Tools Used :

Pylint was used to check code quality, Bandit to look for common Python security problems, and Flake8 to check style and common coding errors.

3. Issues Found and Fixes :

Issue	What was wrong	What I changed
eval() usage	It can run expressions that should not be trusted.	Used an AST-based evaluator that allows only basic arithmetic.
Password in code	The supervisor password was written directly in the program.	Read it from the WAREHOUSE_SUPERVISOR_PASSWORD environment variable.
MD5 hash	Bandit reported the use of a weak hash.	Changed the checksum to SHA-256.
Mutable default list	A default list can be shared between calls.	Changed the history default to None.
Bare exception	Errors could be ignored without explaining what happened.	Removed silent exception handling and added validation.
File operations	Files should close reliably after operations.	Used with statements and UTF-8 encoding.
Style and unused names	Some imports, variables and naming choices were unnecessary or inconsistent.	Removed unused names and improved formatting and function naming.

4. Testing and Results :

Commands used:

```
python .\warehouse_stock_manager.py

python -m pylint .\warehouse_stock_manager.py

python -m bandit -r .\warehouse_stock_manager.py

python -m flake8 .\warehouse_stock_manager.py
```

The program ran successfully. When the supervisor password was set in the PowerShell session, the adjustment was applied and WID-002 changed from 4 to 29 units. The final checks showed Pylint 10.00/10, Bandit reported no issues identified, and Flake8 printed no violations.

5. Reflection :

1. What did you learn from this lab?
I learned that static analysis tools can find problems in code without running it. Each tool checks a different area, so using all three gives a better check.

2. Which issues were important?

The important issues were eval(), the password written in the code, MD5 hashing, and the mutable default list. These could affect security or how reliably the program works.

3. How did you check your fixes?

I changed the code and ran the program again. I also reran Pylint, Bandit and Flake8 to see whether any findings remained.

4. How can you use this in future projects?

I can run these tools while coding and before submitting a project. This can help me catch mistakes earlier and keep the code easier to read.

6.GitHub Repository :

https://github.com/PES2UG24CS246/SE-LABS-2026

Lab Folder : LAB 5